
3.4.6 Chemical elements are recycled in ecosystems. Microorganisms play a key role in recycling these elements.

Nutrient cycles	The role of microorganisms in the carbon and nitrogen cycles in sufficient detail to illustrate the processes of saprobial nutrition, ammonification, nitrification, nitrogen fixation and denitrification. (The names of individual species are not required.)
Carbon	The importance of respiration, photosynthesis and human activity in giving rise to short-term fluctuation and long-term change in global carbon dioxide concentration. The roles of carbon dioxide and methane in enhancing the greenhouse effect and bringing about global warming. Candidates should be able to analyse, interpret and evaluate data relating to evidence of global warming and its effects on • the yield of crop plants • the life-cycles and numbers of insect pests • the distribution and numbers of wild animals and plants.
Nitrogen	The environmental issues arising from the use of fertilisers. Leaching and eutrophication. Candidates should be able to analyse, interpret and evaluate data relating to eutrophication.